

12 Oct. Quiz 2 | Rmks (4.3)

Recall Linear $T: V \rightarrow V$ for f.d. V

invertible

$\Leftrightarrow T$ is bijective

$\Leftrightarrow T$ is injective

$\Leftrightarrow T$ is surjective

$$\dim(V) = \dim(\text{Im } f) + \dim(\ker f)$$

$$\Leftrightarrow \det(T) \neq 0$$

Let U, W subspaces of V , finite dim

$$U \oplus W = V \Leftrightarrow \begin{cases} U \cap W = 0 & \text{"indep"} \\ U + W = V & \text{"span"} \end{cases}$$

$$\begin{array}{l} \text{Rmk} \\ \Leftrightarrow \\ V = U \oplus W \end{array} \left\{ \begin{array}{l} \dim U + \dim W = \dim V \\ U \cap W = 0 \\ U + W = V \end{array} \right\} \begin{array}{l} \text{Any two of these} \\ \text{implies the third} \\ \text{use} \end{array}$$

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

See Artin prop 4.3.1

Reminder, change of basis for an operator "PAP"

Field F $(F, +, 0, \cdot, 1)$ with well behaved four arithmetic opⁿ.

Examples ① Subfields of $\mathbb{C}$ e.g. $\mathbb{C} \supset \mathbb{R} \supset \mathbb{Q}[\sqrt{n}] \supset \mathbb{Q}$
(under inherited opⁿ)
 $\mathbb{Q}[i] = \{a + b\sqrt{n} \mid a, b \in \mathbb{Q}\}$

② Finite fields e.g. $\mathbb{F}_p = \text{set of integers mod } p$ (p a prime) n fixed $\in \mathbb{Z}_{\geq 0}$

Rmk Integers mod n form a "ring" (i.e. have well-behaved $+$, $-$, $\cdot$)

Suppose $\gcd(k, n) = d > 1$ then $\bar{k}$ cannot have an inverse mod n . b/c if $\bar{k}\bar{x} = \bar{1}$

Suppose $\gcd(k, n) = 1$ Does k have an inverse mod n ?

Yes! b/c by Bezout's lemma $\gcd(k, n) = ak + bn \stackrel{\text{here}}{=} 1 \Rightarrow \bar{a}\bar{k} = \bar{1}$ | Multiply by $(\frac{n}{d})$ to get $(\frac{n}{d})\bar{k}\bar{x} = \bar{0} = (\frac{n}{d}) \rightarrow \leftarrow$

Upshot Integers modulo a prime p form a field, denoted $\mathbb{F}_p = \{\bar{0}, \dots, \overline{p-1}\}$

(3) Given any field F , $F(x) = \left\{ \frac{p(x)}{q(x)} \mid \begin{array}{l} p, q \text{ polynomials with coeff in } F \\ q \neq 0 \end{array} \right\}$ is a field.

"Linear algebra 0" is valid over any field. More in tutorial/HW eg $|GL_2(\mathbb{F}_p)| = ?$

(where $GL_n(F)$ = set of
all $n \times n$ invertible matrices
with entries in field F)

$SL_n(F) = \left\{ \begin{array}{l} \text{set of matrices with} \\ \text{determinant 1} \end{array} \right\}$

Towards eigenanalysis 4.4-4.6

Key question: For a given operator

$T: \text{f.dim } V \rightarrow V$, how to find a "good basis" of V s.t. matrix of T wrt that basis is as "nice/simple" as possible

Nice = diagonal, if possible.

Often possible, but not always

$E: T(v) = Nv$, where $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

is not diagonalizable
i.e. $P^{-1}NP$ cannot be a diagonal matrix

Q1 How to decide if an operator T is diagonalizable?

Q2 If T is diagonalizable, how to find a magic basis

Q3 If T is not diagonalizable, what can we do?

Diagonalizing a matrix A means finding a (change of basis matrix) P

s.t. $P^{-1}AP = D$ diagonal matrix
 $\begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$

$\Leftrightarrow$
 $\textcircled{*} AP = PD$

let $P = [w_1 \dots w_n]$ col vectors
Then $\textcircled{*}$ becomes

$$Aw_1 = d_1 w_1, \dots, Aw_n = d_n w_n$$